

# Technical Requirement Document

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Here is the Technical Requirement Document (TRD) based on the provided Functional Requirement Document (FRD):

**Technical Requirement ID: TR-DLT-001**

**Related Functional Requirement(s): FR-DLT-001, FR-DLT-002**

**Objective: Implement a Delta Live Tables (DLT) pipeline to load customer and order data from CSV files into delta tables, perform data cleansing, and create aggregated tables.**

**Target Cluster Configuration:**

**Cluster Name: Gen AI POC Databricks Cluster**

**Databricks Runtime Version: Latest LTS version (e.g., 10.4.x-scala2.12)**

**Node Type: Standard\_DS3\_v2**

**Driver Node: Standard\_DS3\_v2**

**Worker Nodes: 2-5 nodes, depending on the data volume**

**Autoscaling: Enabled**

**Auto Termination: Enabled (e.g., 30 minutes)**

**Libraries Installed: delta-lake, pyspark**

**Source Data Details:**

**Dataset Name: Customer Data**

**Location (Path/Table): `/Volumes/gen\_ai\_poc\_databrickscoe/sdlc\_wizard/customerdata`**

**Format: CSV**

**Description: Customer data with schema: CustId, Name, EmailId, Region**

**Dataset Name: Order Data**

**Location (Path/Table): `/Volumes/gen\_ai\_poc\_databrickscoe/sdlc\_wizard/orderdata`**

**Format: CSV**

**Description: Order data with schema: OrderId, ItemName, PricePerUnit, Qty, Date, CustId**

**Target Data Details:**

**Output Name: Customer Delta Table**

**Location: `gen\_ai\_poc\_databrickscoe.sdlc\_wizard.customer`**

**Format: Delta**

**Description: Customer data delta table**

**Output Name: Order Delta Table**

**Location: `gen\_ai\_poc\_databrickscoe.sdmc\_wizard.order`**

**Format: Delta**

**Description: Order data delta table**

**Output Name: Order Summary Delta Table**

**Location: `gen\_ai\_poc\_databrickscoe.sdmc\_wizard.ordersummary`**

**Format: Delta**

**Description: Joined customer and order data delta table**

**Output Name: Customer Aggregate Spend Delta Table**

**Location: `gen\_ai\_poc\_databrickscoe.sdmc\_wizard.customeraggregatespend`**

**Format: Delta**

**Description: Aggregated customer spend data delta table**

**Job Flow / Pipeline Stages:**

- Read customer and order data from CSV files
- Create delta tables for customer and order data
- Add TotalAmount column to order delta table
- Remove null and duplicate records from customer and order delta tables
- Join customer and order delta tables and load into ordersummary delta table
- Aggregate data from ordersummary delta table and load into customeraggregatespend delta table
- Implement SCD type 2 logic for ordersummary delta table

**Data Transformations / Business Logic:**

**Step: Add TotalAmount column to order delta table**

**Description: Calculate TotalAmount as the product of PricePerUnit and Qty**

**Transformation Logic: `TotalAmount = PricePerUnit Qty`**

**Step: Remove null and duplicate records from customer and order delta tables**

**Description: Remove null and duplicate records to ensure data quality**

**Transformation Logic: Use `dropDuplicates()` and `dropna()` functions**

**Step: Join customer and order delta tables**

**Description: Join customer and order data on CustId**

**Transformation Logic: Use ``join()`` function with inner join**

**Error Handling and Logging:**

- Log errors and exceptions to a designated log location (e.g., ``/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/logs``)
- Implement retry mechanism for failed tasks (e.g., 3 retries with 5-minute interval)
- Use Databricks logging and monitoring features to track job performance and errors